

Performance Analysis of **Federated Learning** under Non-IID Financial Tabular Data Distributions



Submitted by

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PRESENTATION OUTLINE

01 Introduction & Background

02 Problem Statement & Objectives

03 Methodology & Experimental Design

04 Results – IID & Label Skew

05 Results – Quantity Skew

06 FedAvg vs FedProx Comparison

07 Explainability & Practical Implications

08 Conclusion & Future Work

INTRODUCTION & BACKGROUND



Data Privacy Challenge

Digital financial systems generate massive sensitive transactional data. Centralized ML requires data sharing — raising privacy, compliance & confidentiality concerns.



Federated Learning

FL enables collaborative training without sharing raw data. Each client trains locally, sharing only model parameters with a central server (McMahan et al., 2017).



Non-IID Challenge

Real-world FL clients have heterogeneous datasets. Class imbalances, feature differences and size variations violate the IID assumption, degrading convergence.

PROBLEM STATEMENT & RESEARCH OBJECTIVES

PROBLEM STATEMENT

- Heterogeneous datasets across FL clients violate the IID assumption
- Label skew → conflicting gradient updates → slow convergence
- Client drift: local models diverge from global optimization objective
- Performance gap between federated and centralized learning
- Limited systematic study of FL under controlled heterogeneity in finance

RESEARCH OBJECTIVES

- 1 Establish a centralized ML baseline for credit default prediction
- 2 Implement FL under IID distribution and compare with centralized
- 3 Evaluate impact of label skew (70% and 90%) on FL performance
- 4 Examine quantity skew effects on convergence
- 5 Compare FedAvg vs FedProx under Non-IID conditions

METHODOLOGY & EXPERIMENTAL DESIGN

DATASET & CONFIGURATION

Dataset	UCI Credit Card Default (30,000 samples, 24 features)
Split	80–20 (24,000 train / 6,000 test), stratified
Model	Logistic Regression (SGDClassifier, logistic loss)
Clients	10 federated clients
Rounds	40 communication rounds
Local Epochs	10 per round
Learning Rate	0.0005 (constant)
Platform	Python: Scikit-learn, NumPy, Pandas Apple M1

DATA PARTITIONING SCENARIOS

IID

Equal random split (2,400 samples/client). Balanced class distribution. Ideal federated environment.

Label Skew 70%

70% samples from one dominant class per client. Introduces moderate statistical heterogeneity.

Label Skew 90%

90% samples from one class per client. Severe heterogeneity — strong client drift expected.

Quantity Skew

2 clients: 8,000 samples each. 8 clients: 1,000 samples each. Class distribution remains balanced.

RESULTS – CENTRALIZED BASELINE & IID FEDERATED

CENTRALIZED BASELINE (LOGISTIC REGRESSION)

Accuracy 80.72%



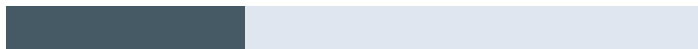
Precision 69.41%



Recall 22.91%



F1-Score 34.45%



⚠ Low recall (22.91%) due to class imbalance — model misses 1,023 actual defaults

FEDERATED LEARNING — IID DISTRIBUTION

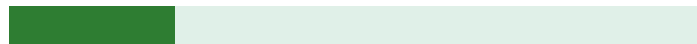
Accuracy 80.80%



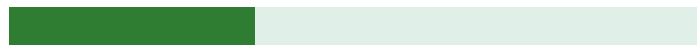
Precision 68.98%



Recall 23.96%



F1-Score 35.57%

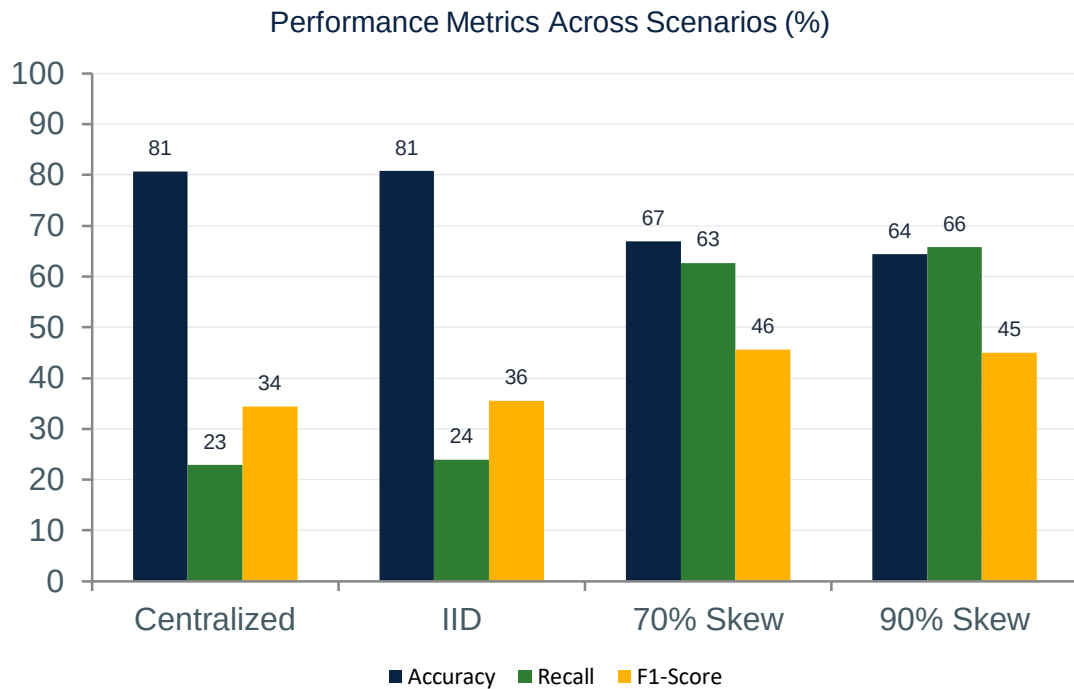


Log Loss 0.4710



✓ Converges in ~10–15 rounds. Matches centralized performance — IID FL is viable!

RESULTS – IMPACT OF LABEL SKEW



KEY OBSERVATIONS

- ↓ Accuracy drops from 80.8% (IID) → 66.9% (70% skew) → 64.4% (90% skew)
- ↑ Recall increases: 23.96% → 62.70% → 65.79% — model favors default class detection
- ⚡ Label skew causes client drift — local gradients conflict during FedAvg aggregation
- Model stabilizes in ~10–15 rounds even under skew, but at lower accuracy

Label skew significantly degrades federated accuracy due to heterogeneous class distributions producing conflicting local gradients — a key challenge for financial FL systems.

RESULTS – QUANTITY SKEW ANALYSIS

Scenario	Accuracy	Precision	Recall	F1-Score	Log Loss
Centralized	0.8072	0.6941	0.2291	0.3445	—
FL – IID	0.8080	0.6898	0.2396	0.3557	0.4710
FL – 70% Label Skew	0.6693	0.3585	0.6270	0.4561	0.6107
FL – 90% Label Skew	0.6438	0.3415	0.6579	0.4497	0.6170
FL – Quantity Skew	0.8080	0.6993	0.2313	0.3477	0.4711

Quantity Skew $\approx$ IID

Despite 2 clients having 8 $\times$ more data (8,000 vs 1,000 samples), accuracy remains at 80.80%. Class distribution is balanced, so gradients align well during FedAvg aggregation.

Label Skew is Damaging

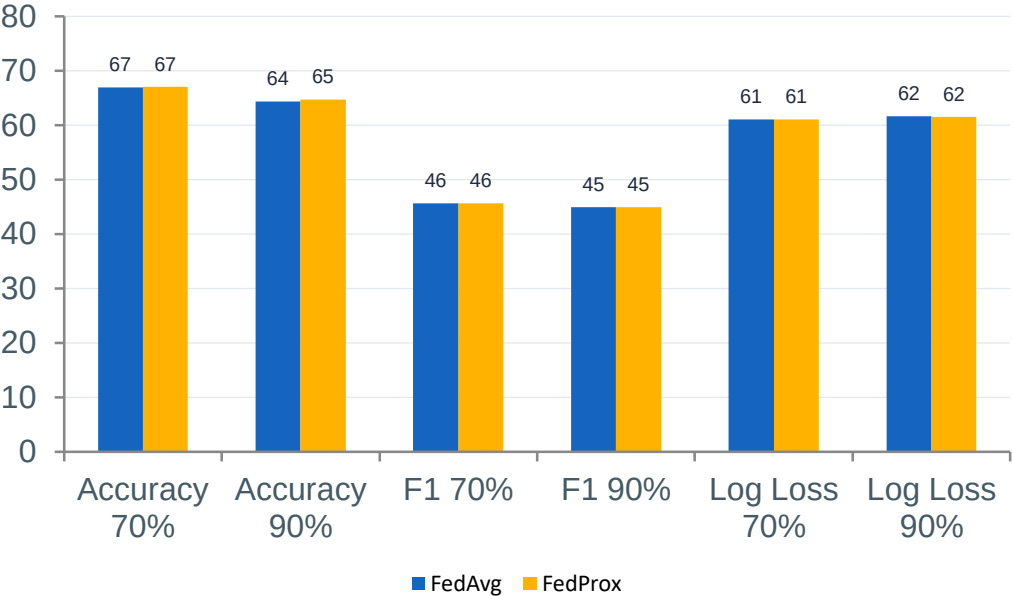
Label skew directly alters the learning objective at each client. Conflicting class distributions produce divergent updates — the root cause of performance degradation.

Key Distinction

Quantity skew affects the magnitude of client contributions. Label skew affects the direction of optimization. The latter is far more harmful for FL convergence.

COMPARATIVE ANALYSIS — FedAvg vs FedProx

FedAvg vs FedProx — Key Metrics (%)



ALGORITHM CHARACTERISTICS

Strategy	Weighted avg.	Proximal reg.
Non-IID handling	Moderate	Better
Complexity	Lower	Slightly higher
Severe skew stability	Lower	Higher
Convergence 70%	~10–15 rounds	~10–15 rounds
Performance gain	Baseline	Marginal ↑

FedProx outperforms FedAvg under **severe label skew (90%)** — higher accuracy (64.78% vs 64.38%) and lower log loss. However, gains are modest with logistic regression, suggesting need for advanced optimization with complex models.

EXPLAINABILITY & PRACTICAL IMPLICATIONS

EXPLAINABILITY (XAI) IN FEDERATED CREDIT RISK

Why Needed?

Financial institutions must justify automated lending decisions to customers, auditors, and regulators — interpretability is mandatory.

LIME & SHAP

Post-hoc explanation methods can identify which features (repayment status, credit utilization, billing amounts) drive default predictions in the global federated model.

Heterogeneity Insight

Comparing feature importance across IID vs Non-IID scenarios can reveal how client drift affects which predictors the global model relies on.

Future Integration

Applying LIME/SHAP to this study's federated models is a direct extension — enabling performance analysis + interpretability for regulatory compliance.

PRACTICAL IMPLICATIONS FOR FINANCIAL INSTITUTIONS



Banks with similar customer distributions are best suited for federated training. Evaluate distributional characteristics before deployment.



Performance–privacy trade-off is real: centralized always outperforms FL, but FL preserves compliance & data ownership.



Severe heterogeneity warrants personalized FL, adaptive weighting, or transfer learning beyond FedAvg/FedProx.



FL is viable for privacy-preserving credit risk prediction when data distributions across partner institutions are not excessively skewed.

CONCLUSION & FUTURE WORK

1

IID Performance Match

FL (FedAvg) achieves 80.80% accuracy under IID — virtually identical to centralized (80.72%). Federated training works when client datasets are statistically similar.

2

Label Skew is the Critical Challenge

Accuracy drops to 66.93% (70% skew) and 64.38% (90% skew). Label heterogeneity causes client drift and conflicting gradients — the primary obstacle in financial FL.

3

FedProx Improves Robustness

Under severe skew, FedProx achieves 64.78% accuracy (vs 64.38% FedAvg) and lower log loss. Proximal regularization constrains local drift — more effective at high heterogeneity.

FUTURE DIRECTIONS

- Advanced models (neural networks, gradient boosting) to reduce performance gap
- SCAFFOLD, FedNova, personalized FL for heterogeneous settings
- Feature distribution shift, concept drift, temporal variation in Non-IID scenarios
- Differential privacy + secure aggregation for stronger privacy guarantees
- LIME / SHAP integration for explainable federated credit risk decisions

THANK YOU

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Open for Questions